

Q(1) The solution of the (separable) D. Equation $\frac{dy}{dx} = (1+y^2)(1-x^2)^{-\frac{1}{2}}$, $y(1) = 0$ is in the form

- a) $\tan^{-1} y = \sin^{-1} x + \frac{\pi}{2}$ **b) $\tan^{-1} y = \sin^{-1} x - \frac{\pi}{2}$** c) $\tan^{-1} y = \sin^{-1} x - \frac{\pi}{6}$ d) $5x^2 + c$

Q(2) The solution of the D. Equation $y' = (y+4x)^2$ is

- a) $\frac{1}{2} \tan^{-1}(y+4x) + x = c$ **b) $\frac{1}{4} \tan^{-1}\left(\frac{y+4x}{2}\right) + x = c$**
c) $\tan^{-1}\left(\frac{y+4x}{2}\right) + x = c$ **d) $\frac{1}{2} \tan^{-1}\left(\frac{y+4x}{2}\right) + x = c$**

Q(3) The differential equation $\cos(x+y)dx + (2y + 3y^2 + \cos(x+y))dy = 0$ is

- a) Homogeneous ~~b) Separable~~ **c) Exact** ~~d) Linear~~

Q(4) The value of the constant c that makes $(9x^2 - cy)dx + (5x + 4y^2)dy = 0$ exact is

- a) -5** b) 5 c) -15 d) 15 e) none

Q(5) The integrating factor for $(2x^2 + y)dx + (x^2y - x)dy = 0$ is

- a) x^{-2}** b) x^2 c) x^{-2} d) x

$$u y' = y^2$$

Q(6) The substitution $u = yx^{-1}$ transforms the equation $xyy' = y^2 + x^2$ into

- ~~a) Linear~~ **b) Homogeneous** c) Separable **d) Exact** e) none

Q(7) Every separable D. Equation is

- a) Linear b) Homogeneous **c) Separable** **d) Exact** e) none

Q(8) The solution of the linear equation $x^2y' + 2xy - x + 1 = 0$, $y(1) = 0$

- a) $\frac{1}{2} - \frac{1}{x} - \frac{x^{-2}}{2}$ **b) $1 - \frac{1}{x} + \frac{x^{-2}}{2}$** **c) $\frac{1}{2} - \frac{1}{x} + \frac{x^{-2}}{2}$** d) $\frac{1}{x} + \frac{x^{-2}}{2}$

Q(9) The solution of the equation $\frac{dy}{dx} - 5y + \frac{5}{2}y^3 = 0$ is

- a) $y^2 = \frac{2}{1 + 2ce^{-10x}}$** b) $y^2 = \frac{3}{1 + 2ce^{-10x}}$ c) $y = \frac{2}{1 + 2ce^{-10x}}$ d) $y^2 = \frac{2}{2 + 2ce^{-10x}}$

Q(10) The D.equation $(x^2 - y^2) dx + xy dy = 0$ is

- a) Exact ☒ b) Homogeneous c) Linear d) separable e) none

Q(11) The substitution $x = u + 2, y = v + 1$ transforms $(x + 2y - 4) dx - (2x + y - 5) dy = 0$ into

- a) exact ☒ b) homogeneous c) separable d) Linear

Q(12) The substitution $u = y^{1-n}$ transforms the equation $y' + p(x)y = g(x)y^n$ into the form

- ☒ a) $u' + (1-n)p(x)u = (1-n)g(x)$ b) $u' + p(x)u = (1-n)g(x)$
c) $u' + (1-n)p(x)u = (1-n)$ d) $u' + (1-n)p(x)u = (1-n)g(x)$

Q(13) The substitution $z = x + y$ transforms $(x + y + 3) dx + (2x + 2y + 4) dy = 0$ into

- a) exact ☒ b) homogeneous c) separable d) Linear

Q(14) The solution of the D.E. $y' = 6y^2x$ is

- a) $c + 3x^2$ b) $c - 3x^2$ c) $\frac{1}{3x^2}$ ☒ d) $\frac{1}{c-3x^2}$

Q(15) The Homogeneous D.Equation is an extension to the

- a) Linear D.Equ. b) Exact D.Equ. c) Bernoulli D.Equ.

☒ d) Separable D.Equ.

Q₁ ∴ The Solution of the (separable) D.E

$$\frac{dy}{dx} = (1+y^2)(1-x^2)^{-\frac{1}{2}}, \quad y(1)=0$$

Sol: $\int \frac{dy}{1+y^2} = \int \frac{1}{\sqrt{1-x^2}} dx$

$$\tan^{-1} y = \sin^{-1} x + c$$

$$y(1)=0$$

$$0 = \frac{\pi}{2} + c$$

$$-\frac{\pi}{2} = c$$

$$\boxed{\tan^{-1} y = \sin^{-1} x - \frac{\pi}{2}}$$

Q₂ ∴ The Solution of the D.E $y' = (y+4x)^2$ is ∴

Sol: Let $t = y + 4x$

$$y = t - 4x$$

$$\frac{dy}{dx} = \frac{dt}{dx} - 4, \quad \frac{dy}{dx} = t^2$$

$$\frac{dt}{dx} - 4 = t^2$$

$$\frac{dt}{dx} = t^2 + 4$$

$$\int \frac{dt}{t^2+4} = \int dx$$

$$\boxed{\frac{1}{2} \tan^{-1}\left(\frac{t}{2}\right) = x + c}$$

$$\rightarrow \frac{1}{2} \tan^{-1}\left(\frac{y+4x}{2}\right) = x + c$$

Q₃: The D.E $\cos(x+y)dx + (2y+3y^2+\cos(x+y))dy=0$ is:

a) Homogeneous b) Separable c) Exact d) Linear

Sol: $\frac{\partial M}{\partial y} = -\sin(x+y) \quad \left\{ \quad \frac{\partial N}{\partial x} = -\sin(x+y) \right.$

Exact (c)

Q₄: The value of the constant c that makes

$(9x^2 - cy)dx + (5x + 4y^2)dy = 0$ Exact is:

Sol: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow -c = 5$
 $c = -5$

Q₅: The integrating factor for $(2x^2+y)dx + (x^2y-x)dy=0$

Sol: $\frac{\partial M}{\partial y} = 1, \frac{\partial N}{\partial x} = 2xy-1$ Non exact $\Rightarrow \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = R(x)$

$\frac{1 - 2xy + 1}{x^2y - x} = \frac{2 - 2xy}{x^2y - x} = \frac{2(1-xy)}{x(xy-1)} = \frac{-2}{x}$

$M(x) = e^{\int R(x)dx} = e^{\int \frac{-2}{x} dx} = e^{-2 \ln x} = \boxed{x^{-2}}$

Q₆: The substitution $u = yx^{-1}$ transforms the equation $xyy' = y^2 + x^2$ into:

a) Linear b) Homogeneous c) Separable d) Exact e) None

Sol: $y' = \frac{y^2 + x^2}{xy} \quad (\div x^2) \Rightarrow \frac{(\frac{y}{x})^2 + 1}{(\frac{y}{x})}$

$u = \frac{y}{x}$
Hom. (b)

Q₇: Every separable D.E is:

a) Linear b) Homogeneous c) Separable **d) Exact** e) None

Sol: ~~XXXXXXXXXX~~

Q₈: The solution of the linear Equation $x^2 y' + 2xy - x + 1 = 0$, $y(1) = 0$

Sol: $y' + \left(\frac{2}{x}\right)y = \frac{x-1}{x^2}$ $P(x) = \frac{2}{x}$, $Q(x) = \frac{x-1}{x^2}$

$$M(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$$

$$y = \frac{1}{x^2} \left[\int x^2 \left(\frac{x-1}{x^2} \right) dx + c \right]$$

$$y = \frac{1}{x^2} \left[\frac{x^2}{2} - x + c \right]$$

$$y = \frac{1}{2} - \frac{1}{x} + \frac{c}{x^2} \quad \underline{\text{But}} \quad y(1) = 0$$

$$0 = \frac{1}{2} - \frac{1}{1} + c$$

$$0 = -\frac{1}{2} + c$$

$$\boxed{\frac{1}{2} = c}$$

$$\boxed{y = \frac{1}{2} - \frac{1}{x} + \frac{x^{-2}}{2}}$$

Q₉: The solution of the equation $\frac{dy}{dx} - 5y + \frac{5}{2}y^3 = 0$ is:

Sol: $\frac{dy}{dx} - 5y = -\frac{5}{2}y^3$

Let $u = y^{-2}$

$\frac{du}{dx} = -2y^{-3} \frac{dy}{dx}$
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$-2y^{-3} \frac{dy}{dx} + 10y^{-2} = 5$
 $\frac{du}{dx} + 10u = 5$

$\frac{du}{dx} + 10u = 5$

$P(x) = 10, G(x) = 5$
 $\mu(x) = e^{\int 10 dx} = e^{10x}$

$u = \frac{1}{e^{10x}} \left[\int 5e^{10x} dx + c \right]$

$u = \frac{1}{e^{10x}} \left[\frac{e^{10x}}{2} + c \right]$

$u = \frac{1}{2} + ce^{-10x}$

$\frac{1}{y^2} = \frac{1 + 2ce^{-10x}}{2}$

$y^2 = \frac{2}{1 + 2ce^{-10x}}$

$y = \sqrt{\frac{2}{1 + 2ce^{-10x}}}$

Q₁₀: The D.E $(x^2 - y^2)dx + (xy)dy = 0$ is:

a) Exact b) Homogeneous c) linear d) separable e) none

Sol: $(xy)dy = (y^2 - x^2)dx$

$\frac{dy}{dx} = \frac{y^2 - x^2}{xy}$ ($\div x^2$) $\Rightarrow$ Homo (b)

Q₁₁: The substitution $x = u + 2$, $y = v + 1$ transforms

$$(x + 2y - 4) dx - (2x + y - 5) dy = 0 \text{ into}$$

a) Exact b) homogeneous c) separable d) linear

Sol: homo. (b)

Q₁₂: The substitution $u = y^{(1-n)}$ transforms the equation
 $y' + P(x)y = g(x)y^n$ into the form:

Sol: Multiply the equation by $(1-n)y^{-n}$

$$u' + (1-n)P(x)u = (1-n)g(x)$$

$$u = y^{(1-n)} \\ \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

Q₁₃: The substitution $z = x + y$ transforms

$$(x + y + 3) dx + (2x + 2y + 4) dy = 0 \text{ into}$$

a) Exact b) homogenous c) separable d) linear

Sol: Separable (c)

Q₁₄: The solution of the D.E $\frac{dy}{dx} = 6y^2 x$ is:

Sol: $\frac{dy}{6y^2} = x dx \Rightarrow \int \frac{1}{6} y^{-2} dy = \int x dx$

$$-\frac{1}{6} y^{-1} = \frac{x^2}{2} + C$$

$$y = \frac{1}{C - 3x^2}$$

$$y^{-1} = -3x^2 + C \left(\frac{-6}{1} \right) \\ y^{-1} = -3x^2 + C$$

Q₁₅: The homogenous D.E is an extension to the:

a) Linear D.E b) Exact D.E c) Bernoulli D.E d) Separable D.E

Sol: Separable D.E d

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